西湖大學
WESTLAKE UNIVERSITY

Topological Quantum Molecular Dynamics

VISTA

Mar 26, 2025

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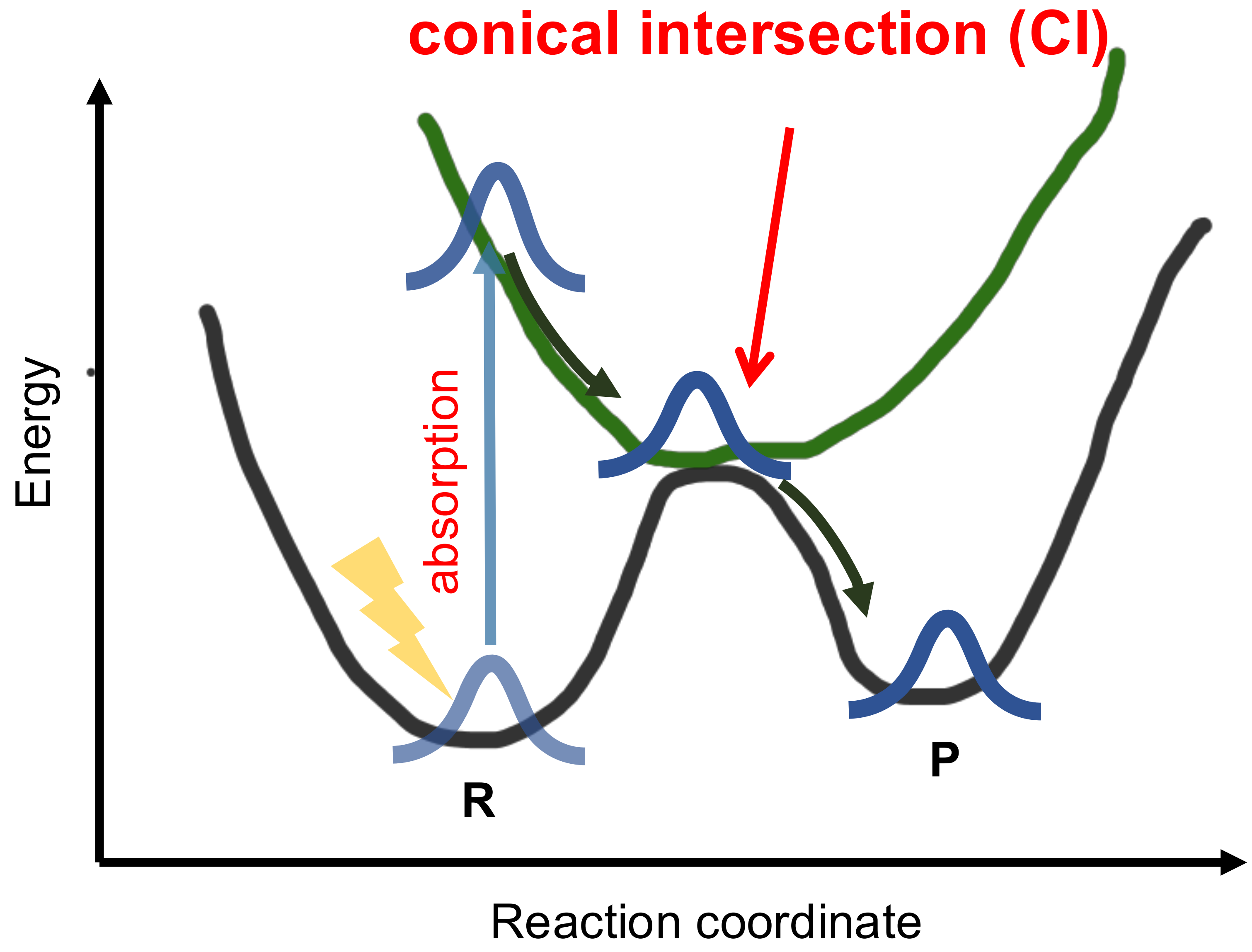
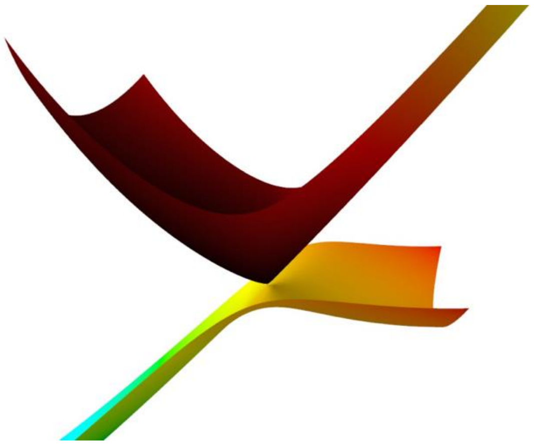
Outline

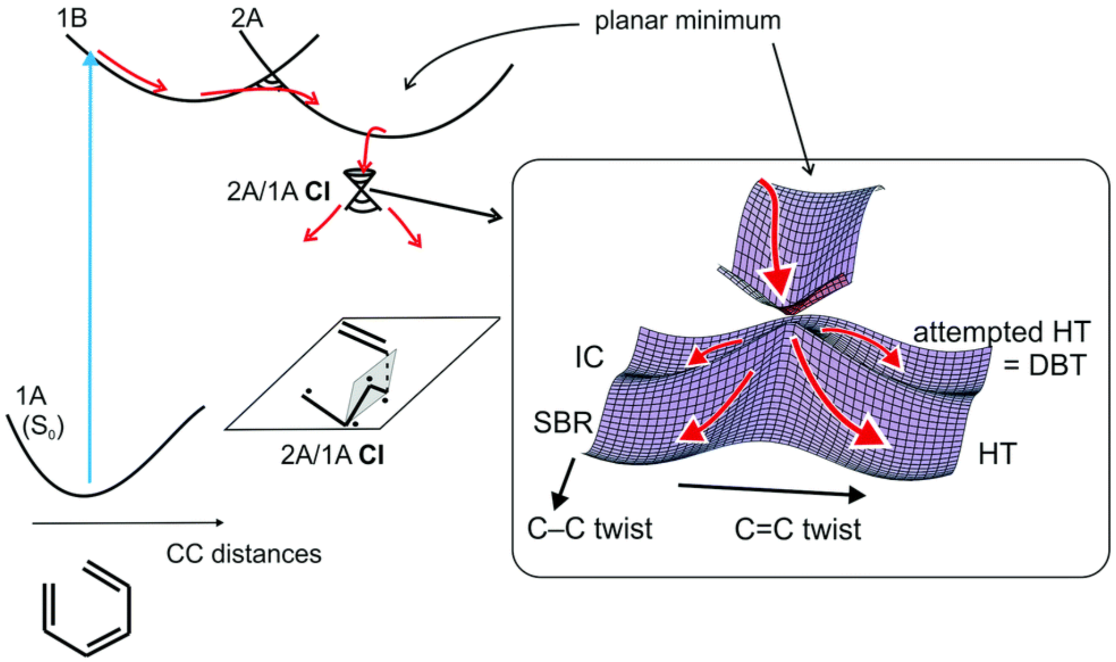
- **Introduction to conical intersections**
- **Challenges of the Born-Oppenheimer framework**
- **A local diabatic ansatz**
- **Topology is the key**

How do molecules behave upon photoexcitation?



Conical intersection: "transition state" for photochemistry

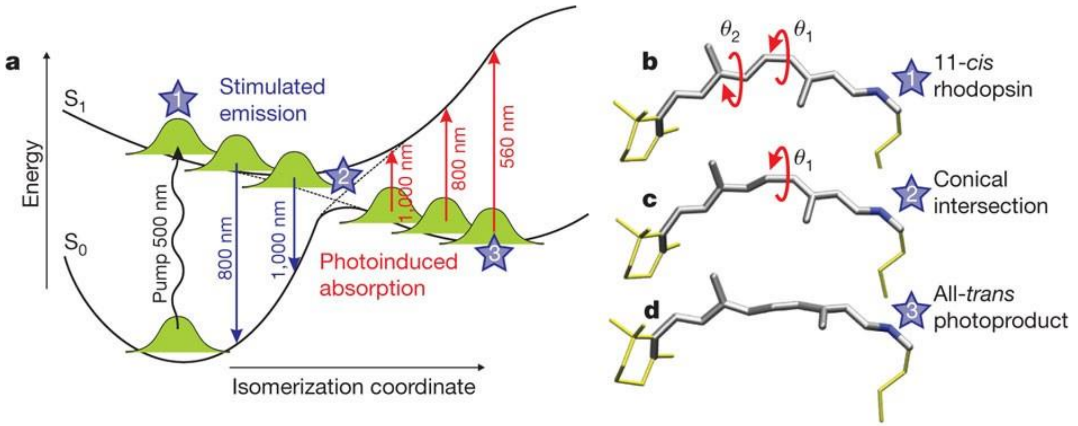




Interaction of two (or more) potential energy surfaces

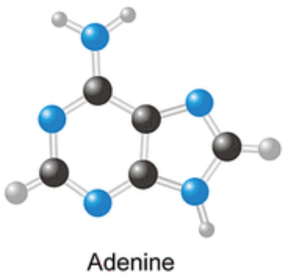
Vision - photoisomerization of rhodopsin

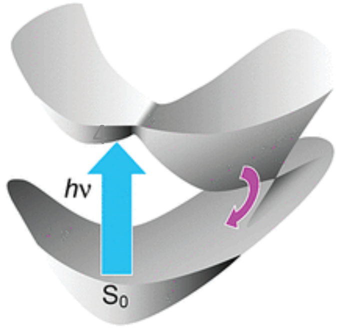
Polli et al. *Nature* 2010, *467* (7314), 440–443



Photostability of DNA molecules

Kang et al. *JACS* **2002**, *124* (44), 12958–12959





Can we monitor conical intersection dynamics by ultrafast spectroscopy?

Imaging CF_3I conical intersection and photodissociation dynamics with ultrafast electron diffraction

Yang et al.,Science 2018, 361,64–67

PHYSICAL REVIEW LETTERS **120,** 243001 (2018)

Ultrafast X-Ray Spectroscopy of Conical Intersections

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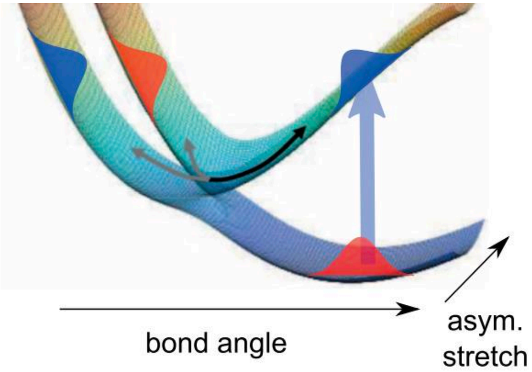
[3]*National Research Council of Canada, 100 Sussex Drive, Ottawa, Ontario K1A 0R6, Canada*
Department of Physics, University of Ottawa, 150 Louis Pasteur, Ottawa, Ontario K1N 6N5, Cana

Conical Intersection Dynamics in NO_2 Probed by Homodyne High-Harmonic Spectroscopy

H. J. Wörner et al., Science 2011, 334, 208



Few-fs resolution of a photoactive protein traversing a conical intersection

https://doi.org/10.1038/s41586-021-04050-9
Received: 12 November 2020

A. Hosseinizadeh[1], N. Breckwoldt[2,3,4], R. Fung[1], R. Sepehr[1], M. Schmidt[1], P. Schwander[1], R. Santra[2,3,4] & A. Ourmazd[1]

THE CROSSING OF POTENTIAL SURFACES[1]

E. TELLER

Department of Physics, George Washington University, Washington, D. C.

Received October 29, 1936

The crossing of potential curves or surfaces is important if reaction mechanisms are discussed, particularly if activated states are involved. Thus in photochemistry both the chemical yield and the fluorescence will depend on the question whether the molecules can get from a higher potential surface to a lower one, transforming in this way electronic excitation energy into kinetic energy of atoms and finally into heat. We know that in both diatomic and polyatomic molecules this can happen in several ways through the change of position of an electron, as discussed by Franck

Studies of the Jahn–Teller effect
II. The dynamical problem

By H. C. Longuet-Higgins,* U. Öpik,† M. H. L. Pryce, F.R.S.‡ and R. A. Sack§

*§*Department of Theoretical Chemistry, University of Cambridge*

†*Department of Applied Mathematics, University College of Wales, Aberystwyth*

‡*Department of Physics, University of Bristol*

(*Received* 14 *August* 1957)

This paper examines the vibronic energy levels of a symmetrical non-linear molecule in a spatially doubly degenerate electronic state which is split in first order by a doubly degenerate vibrational mode. The vibronic levels are classified by a quantum number, which ... to the combined angular momentum of electronic and ... alues are obtained for the energies of these levels as func- ... a dimensionless parameter measuring the magnitude ... g.

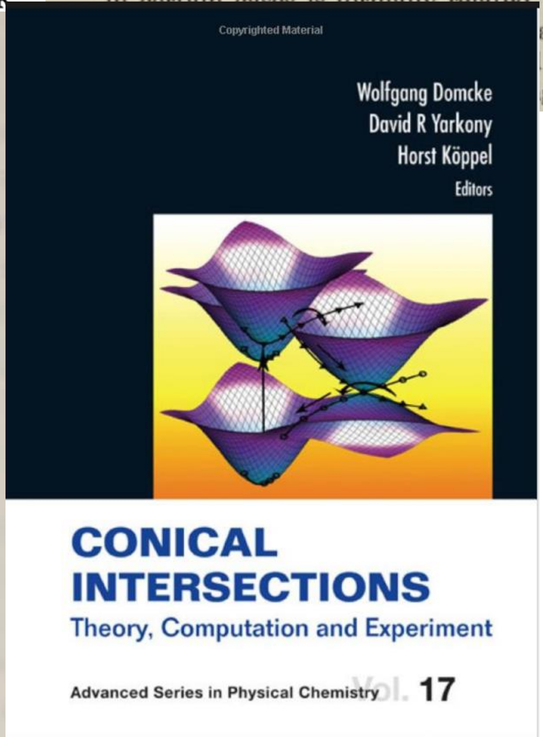
The intersection of potential energy surfaces in polyatomic molecules

By H. C. Longuet-Higgins, F.R.S.

Centre for Research on Perception and Cognition, Laboratory of Experimental Psychology, University of Sussex

(*Received* 5 *December* 1974 – *Revised* 30 *January* 1975)

It is proved that if the wave function of a given electronic state changes sign when transported adiabatically round a loop in nuclear configuration space, then the state must become degenerate with another one at some point within the loop. It is further shown that this condition is satisfied by certain unsymmetrical triatomic systems, thereby disposing of a recent claim that the non-crossing rule for diatomic molecules applies also to polyatomic molecules.

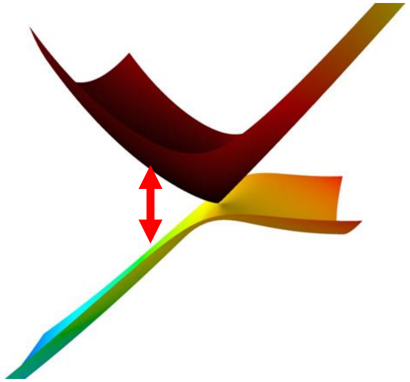


"the rule rather than the exception in polyatomic molecules"

Conical intersection dynamics is very challenging

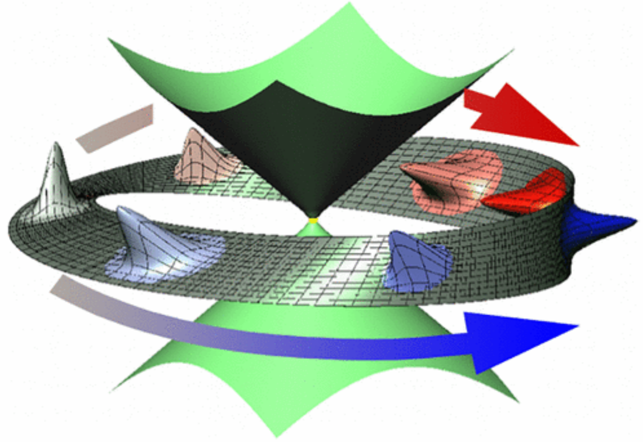
- **Breakdown of Born-Oppenheimer approximation**

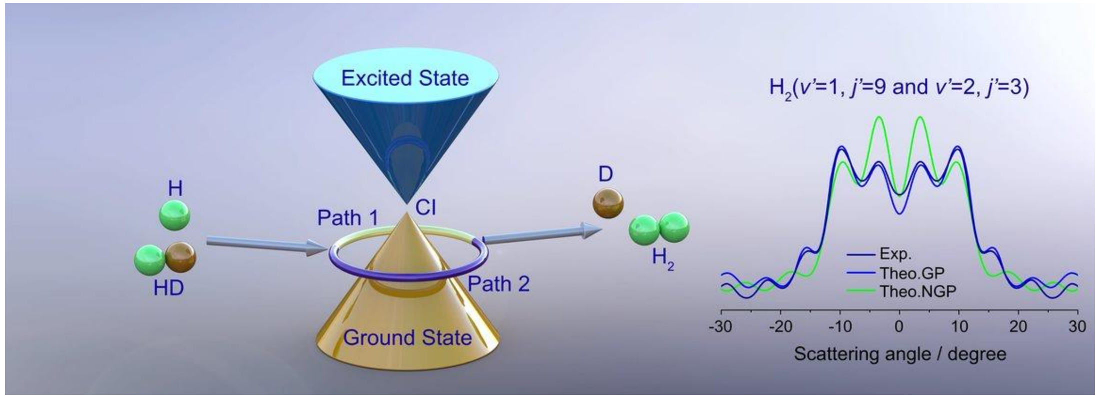
  strongly correlated electron-nuclear dynamics



- **Geometric phase effect**

  neglected in most quantum dynamics methods





Yuan et al. Science **2018**, 362, 1289
Ryabinkin et al. Acc. Chem. Res. **2017**, 50, 7, 1785–1793

The targeted equation

$$\mathrm{i}\partial_t \Psi(\mathbf{r}, \mathbf{R}, t) = H\Psi(\mathbf{r}, \mathbf{R}, t)$$

$$H = \hat{T}_{\mathrm{N}} + H_{\mathrm{BO}}(\mathbf{R})$$

Electronic Hamiltonian

$$H_{\mathrm{BO}}(\mathbf{R}) = \sum_i -\frac{1}{2}\Delta_i + \sum_{i,j} \frac{1}{|r_i - r_j|} + \sum_{I,J} \frac{Z_I Z_J}{|R_i - R_j|} - \sum_{I,i} \frac{Z_I}{|r_i - R_I|}$$

The Born-Huang framework

Born-Huang expansion

nuclear wave packets

$$\Psi(\mathbf{r}, \mathbf{R}, t) = \sum_{\alpha} \underbrace{\phi_\alpha(\mathbf{r}; \mathbf{R})}\overbrace{\chi_\alpha(\mathbf{R}, t)}$$

adiabatic electronic states: eigenstates of the electronic Hamiltonian

$$H_{\mathrm{BO}}(\mathbf{R})\,|\phi_\alpha(\mathbf{R})\rangle = \underbrace{E_\alpha(\mathbf{R})}\,|\phi_\alpha(\mathbf{R})\rangle$$

adiabatic potential energy surface

Born-Oppenheimer dynamics

diagonal Born-Oppenheimer corrections (DBOC)

$$i\partial_t \chi_\alpha(\mathbf{R},t) = \boxed{(T_\mathrm{N} + V_\alpha(\mathbf{R})} + G_{\alpha\alpha}(\mathbf{R}))\chi_\alpha(\mathbf{R},t)$$
$$- \sum_{\beta\neq\alpha} M^{-1}F_{\alpha\beta}\cdot\nabla\chi_\beta(\mathbf{R},t) + \sum_{\beta\neq\alpha} G_{\alpha\beta}\chi_\beta(\mathbf{R},t)$$

first-derivative coupling

second-derivative coupling

$$\text{Born-Oppenheimer} \xrightarrow{F} \xrightarrow{G} \xrightarrow{\text{DBOC}} \longrightarrow \text{Exact}$$

$$\mathbf{F}_\alpha(\mathbf{R}) = \langle\phi_\alpha(\mathbf{R})|\boldsymbol{\nabla}\phi_\alpha(\mathbf{R})\rangle \neq 0$$

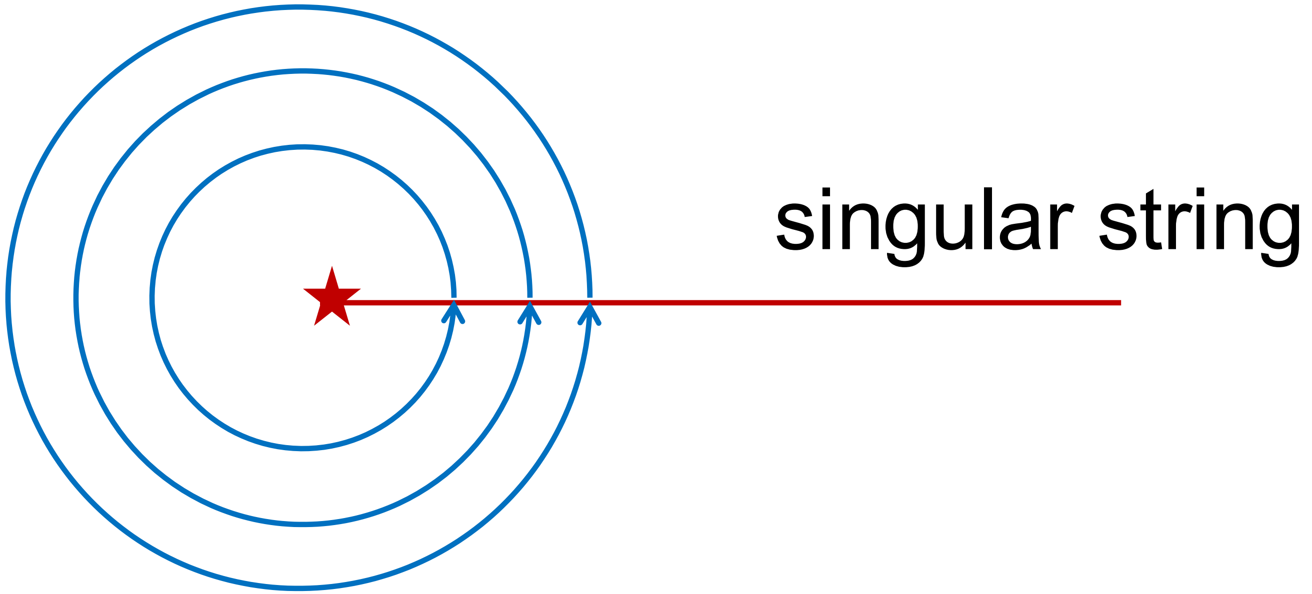
All terms are singular

$$\mathbf{F}_{eg}(\mathbf{R}) = -\frac{\langle \psi_e(\mathbf{R}) | \nabla_R H_{\mathrm{BO}}(\mathbf{R}) | \psi_g(\mathbf{R}) \rangle}{E_e(\mathbf{R}) - E_g(\mathbf{R})}$$

$$G_{eg}(\mathbf{R}) \propto \frac{1}{(E_e(\mathbf{R}) - E_g(\mathbf{R}))^2}$$

Close to conical intersections

$$\mathbf{F}(\mathbf{R}_{\mathrm{CI}}) = \infty \qquad G(\mathbf{R}_{\mathrm{CI}}) = \infty^2$$



$$i\oint F(\mathbf{R})\mathrm{d}\mathbf{R} = \pi$$

$$E_{\mathrm{DBOC}} = \langle \phi | \sum_{\mu} -\frac{1}{2M_{\mu}} \partial_{\mu}^{2} \phi \rangle$$

The diagonal Born–Oppenheimer correction to molecular dynamical properties

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[b] Department of Chemistry, Northwestern University, Evanston, IL 60208, USA

Received 7 September 2000; in final form 18 October 2000

On the inclusion of the diagonal Born-Oppenheimer correction in surface hopping methods

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(Received 18 February 2016; accepted 30 March 2016; published online 15 April 2016)

Wave function continuity and the diagonal Born-Oppenheimer correction at conical intersections

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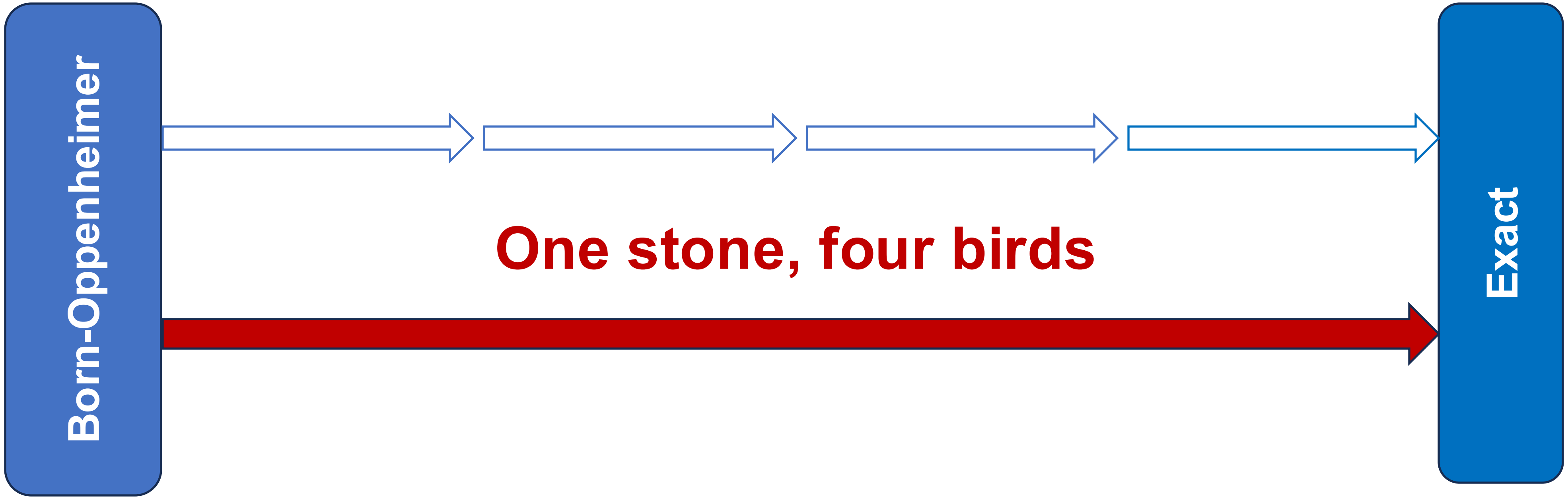
(Received 15 January 2016; accepted 26 April 2016; published online 12 May 2016)

A **topological** framework for understanding and modeling **exact a** **initio** conical intersection dynamics



Y. Xie, R. Liu, B. Gu, *Topological Quantum Molecular Dynamics*, submitted
Y. Xie, B. Gu, arXiv:2501.05003, **2025**
B. Gu, *J. Chem. Theory Comput.* **2024**, 20, 7, 2711–2718
X. Zhu, B. Gu, *J. Phys. Chem. Lett.* **2024**, *15* (33), 8487–8493
Y. Xie, Y. Yang, A. Chen, B. Gu, *J. Chem. Theory Comput.* **2024**, 20 (21), 9512-9521
B. Gu, *J. Chem. Theory Comput.* **2023**, 19, 19, 6557–6563

Remove the nuclear dependence of the electronic states?

$$\boxed{\mathrm{i}\partial_t \Psi(\mathbf{r},\mathbf{R},t) = \left(-\sum_I \frac{1}{2M_I}\frac{\partial^2}{\partial R_I^2} + H_{\mathrm{BO}}(\mathbf{R})\right)\Psi(\mathbf{r},\mathbf{R},t)}$$

$$\Psi(\mathbf{r},\mathbf{R},t) = \sum_\alpha \phi_\alpha(\mathbf{r};\cancel{\mathbf{R}})\chi_\alpha(\mathbf{R},t)$$

Single reference geometry $\phi_\alpha(\mathbf{r};\mathbf{R_0})$

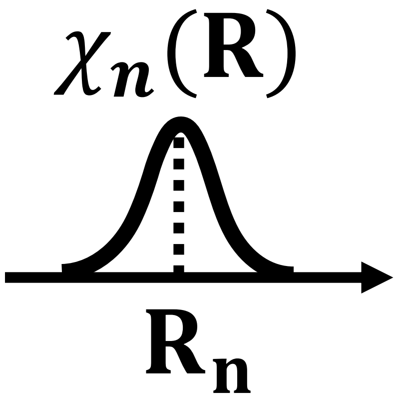
Can we do multiple references? **How do we choose?**

$$\{\mathbf{R}_n\}$$

Choose by discrete variable representation

Choose a primitive nuclear basis set, construct the position eigenstates

$$\hat{\mathbf{R}}\,|\chi_n\rangle = \mathbf{R}_n\,|\chi_n\rangle$$



Our ansatz $$\Psi(\mathbf{r},\mathbf{R},t) = \sum_n \psi_n(\mathbf{r},t)\underbrace{\chi_n(\mathbf{R})}_{\text{localized wave packet centered at } \mathbf{R}_n}$$

Using **adiabatic** electronic states at $\mathbf{R}_n$

$$\psi_n(\mathbf{r},t) = \sum_\alpha C_{n\alpha}(t)\phi_\alpha(\mathbf{r};\mathbf{R}_n)$$

B. Gu, *J. Chem. Theory Comput.* **2023**, 19, 19, 6557−6563

Our ansatz $$\Psi(\mathbf{r},\mathbf{R},t) = \sum_{n,\alpha} C_{n\alpha}(t)\,\phi_\alpha(\mathbf{r};\mathbf{R_n})\chi_n(\mathbf{R})$$

$$H_{\mathrm{BO}}(\hat{\mathbf{R}})|\phi_\alpha(\mathbf{R_n})\rangle|\chi_n\rangle = H_{\mathrm{BO}}(\mathbf{R_n})|\phi_\alpha(\mathbf{R_n})\rangle|\chi_n\rangle$$
$$= V_\alpha(\mathbf{R_n})|\phi_\alpha(\mathbf{R_n})\rangle|\chi_n\rangle$$

$$V_{\beta m} \equiv V_\beta(\mathbf{R}_m)$$

Exact!

$$\boxed{i\dot{C}_{m\beta}(t) = V_{m\beta}C_{m\beta}(t) + T_{mn}A_{m\beta,n\alpha}C_{n\alpha}(t)}$$

Looks just like the Born-Oppenheimer dynamics!

$$\boxed{i\dot{C}_{m\beta}(t) = V_{m\beta}C_{m\beta}(t) + T_{mn}C_{n\beta}(t)}$$

The "Stone": Global electronic overlap matrix

$$A_{m\beta,n\alpha} = \langle \phi_\beta(\mathbf{R}_m) | \phi_\alpha(\mathbf{R}_n) \rangle_{\mathbf{r}}$$

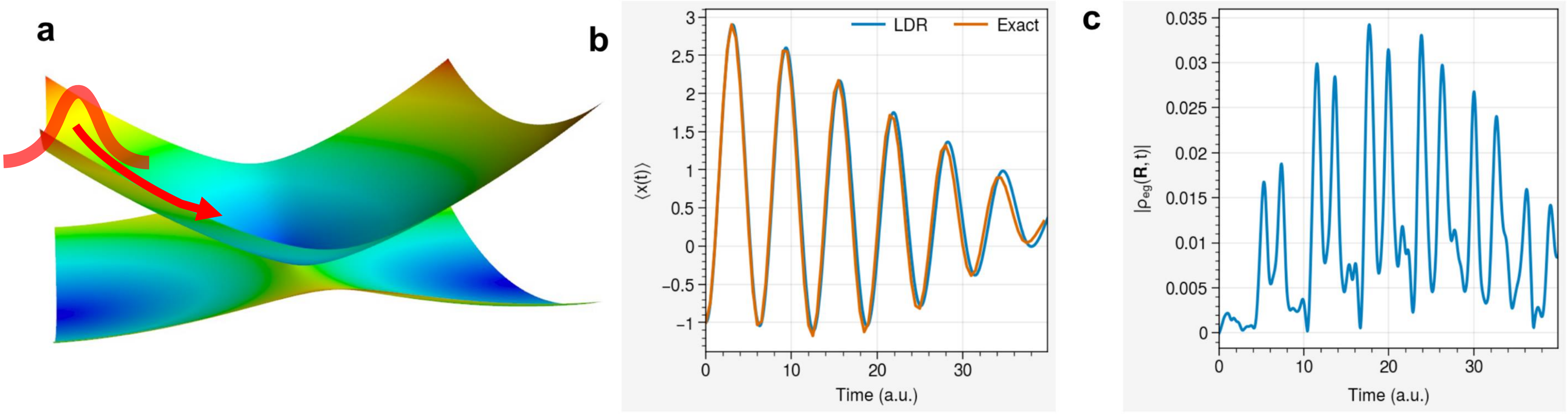
measure how electronic states vary with nuclear geometry

$A_{m\beta,n\alpha} = \delta_{\beta\alpha}$ $\longrightarrow$ Born-Oppenheimer dynamics

- **ALL** effects beyond Born-Oppenheimer unified into a single matrix
- **Divergence free!** Even at the intersection seam

Model 1: Captures electronic transitions & electronic coherence

A two-mode two-state vibronic model



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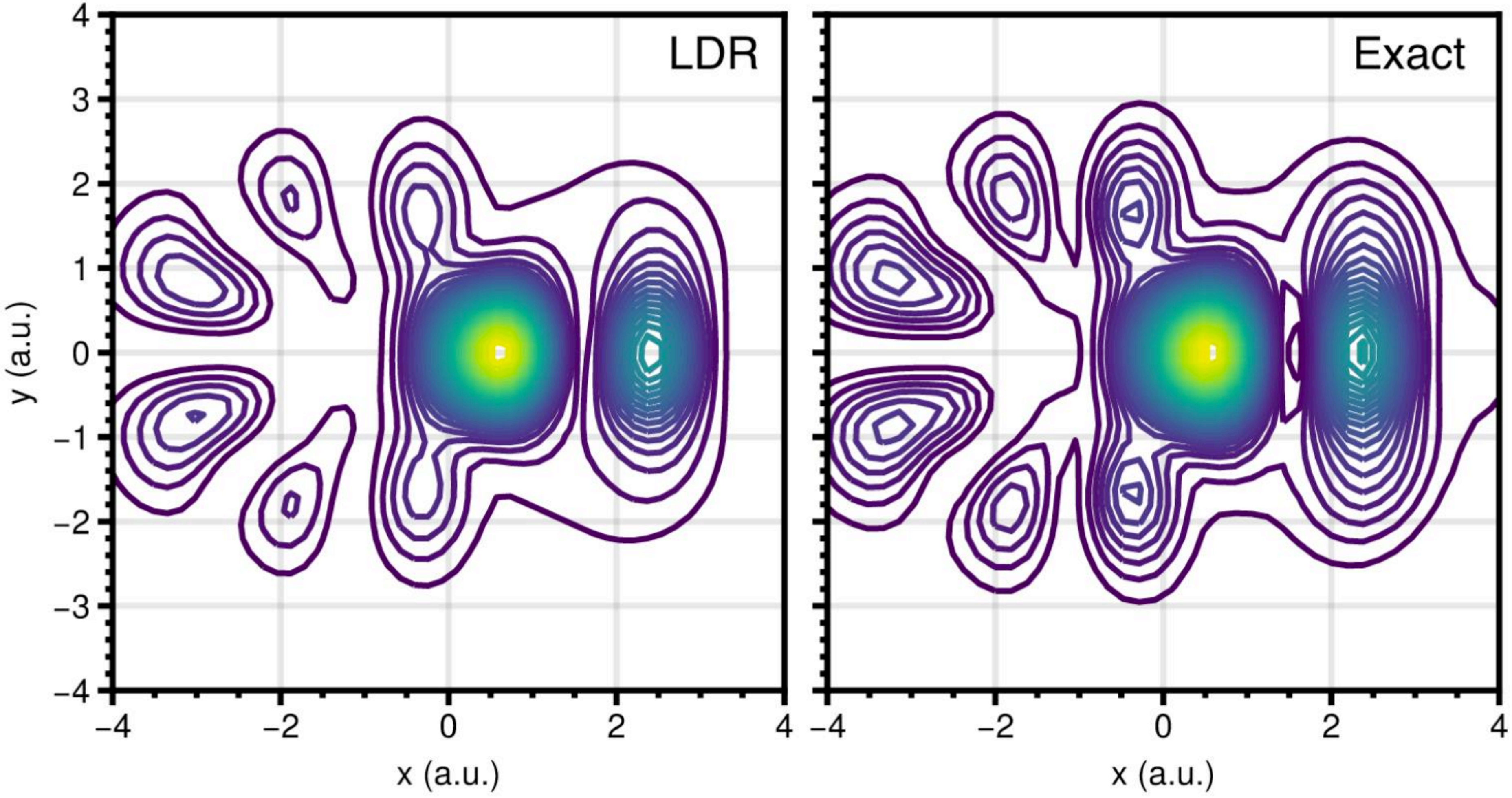
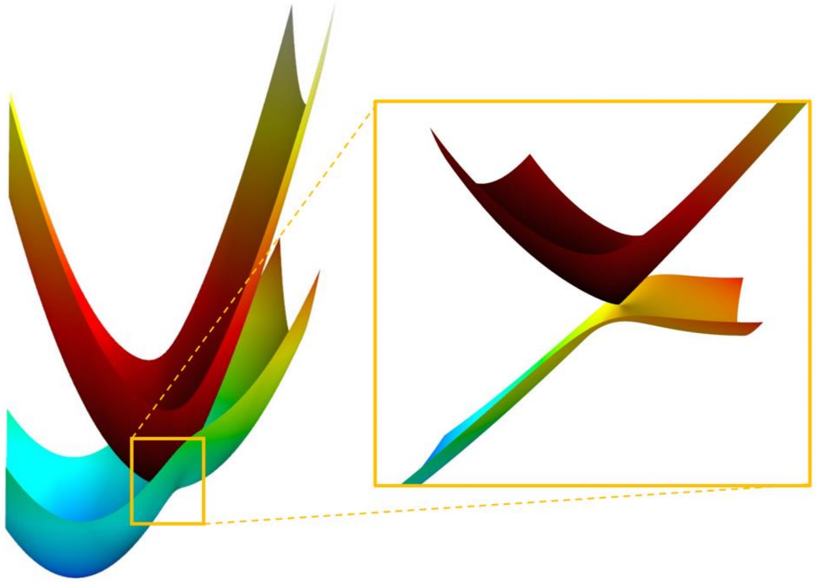
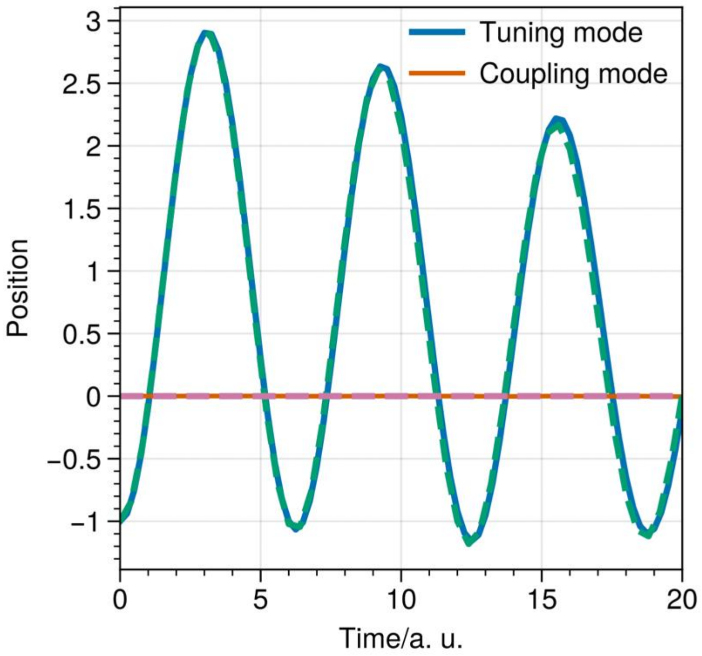
Capture geometric phase

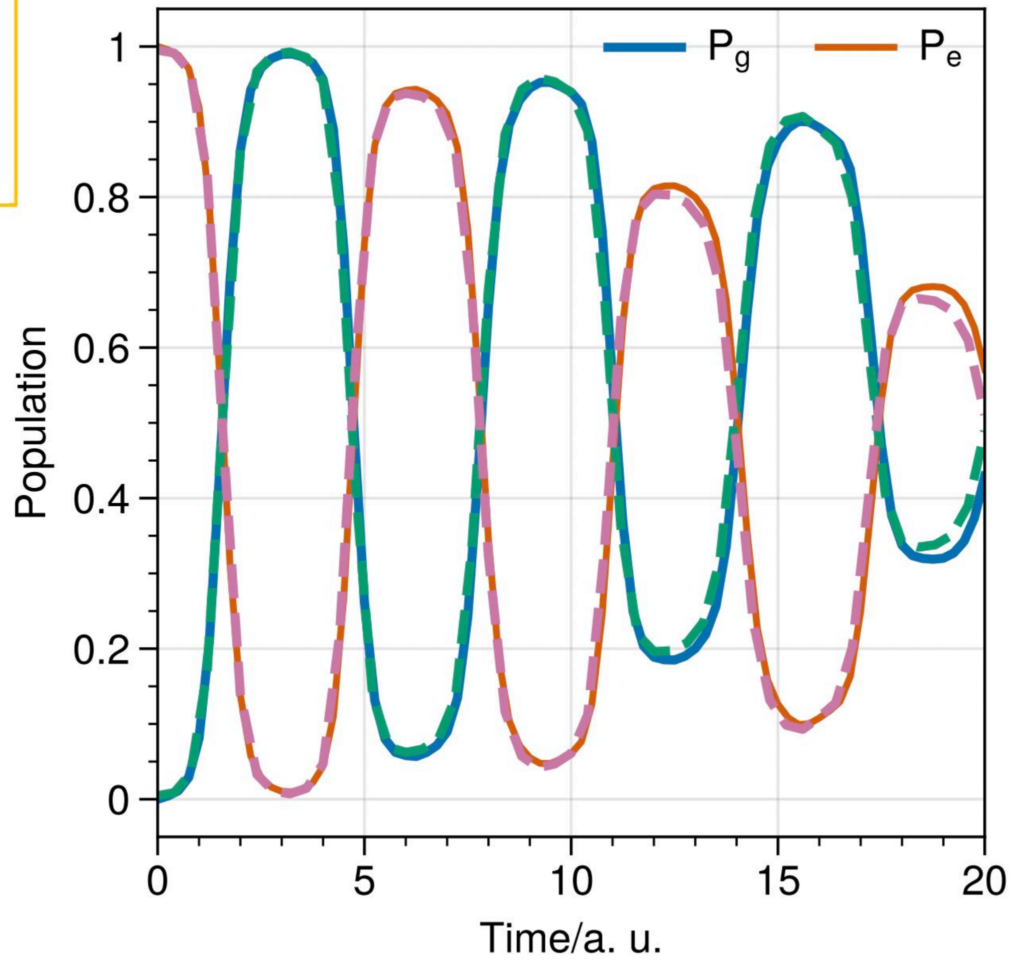


B. Gu, *J. Chem. Theory Comput.* **2023**, 19, 19, 6557−6563

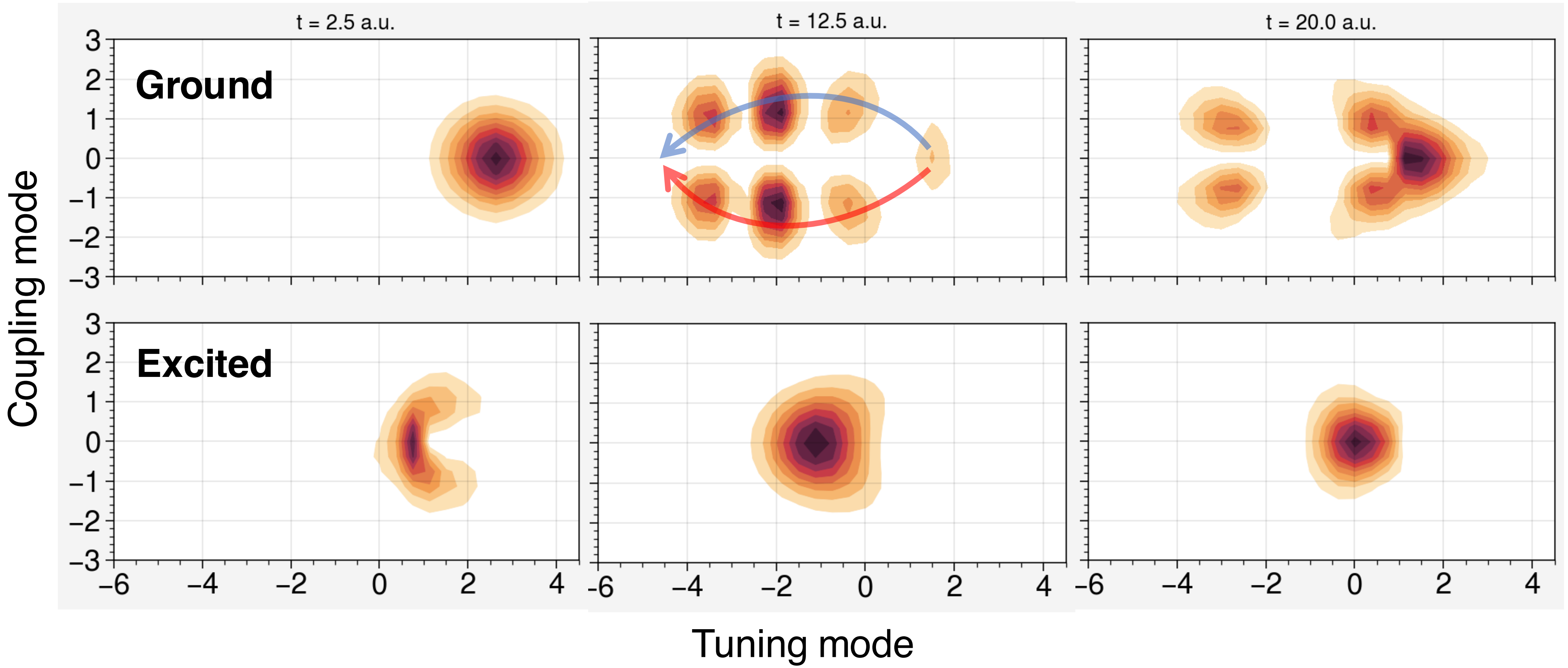
Model 2







B. Gu, *J. Chem. Theory Comput.* **2024**, 20, 7, 2711−2718


t = 2.5 a.u.
t = 12.5 a.u.
t = 20.0 a.u.
Ground
Excited
Coupling mode
Tuning mode
3
2
1
0
−1
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−3
−6
−4
−2
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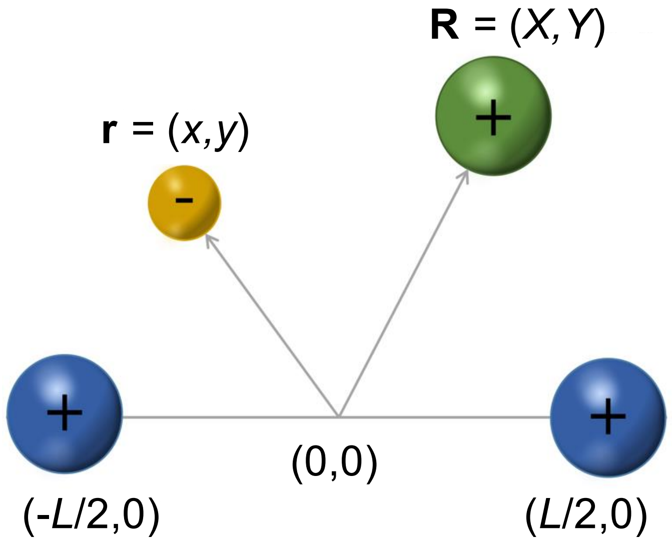
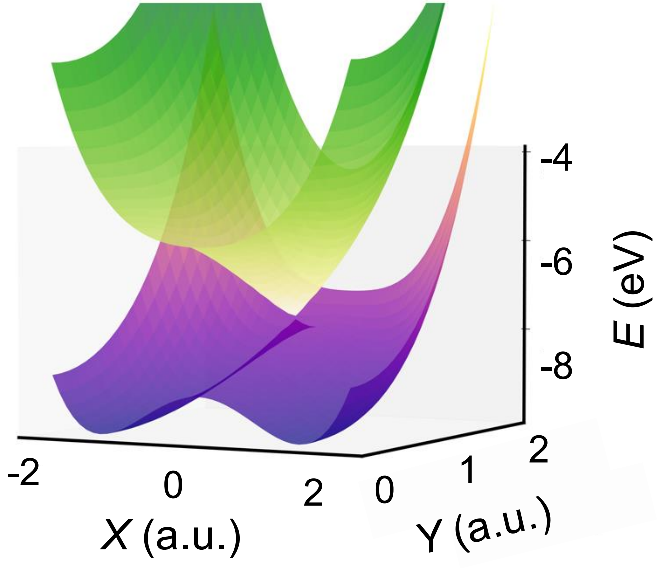
2
4

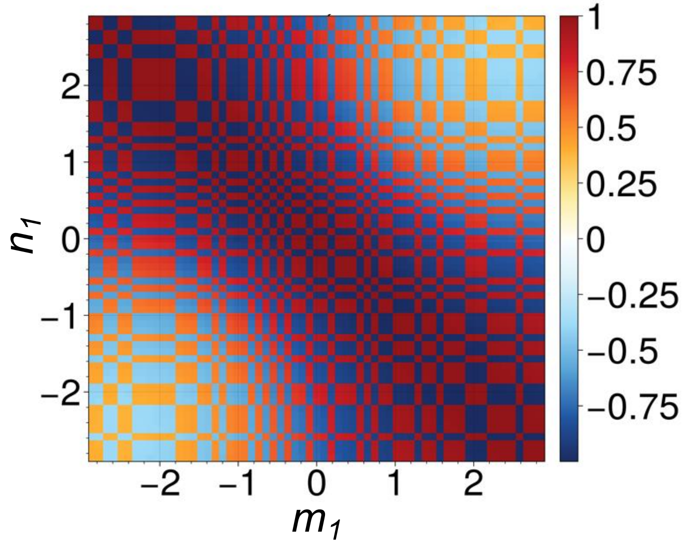
Model 3: Proton-coupled electron transfer

Combined with electronic structure

Xiaotong Zhu

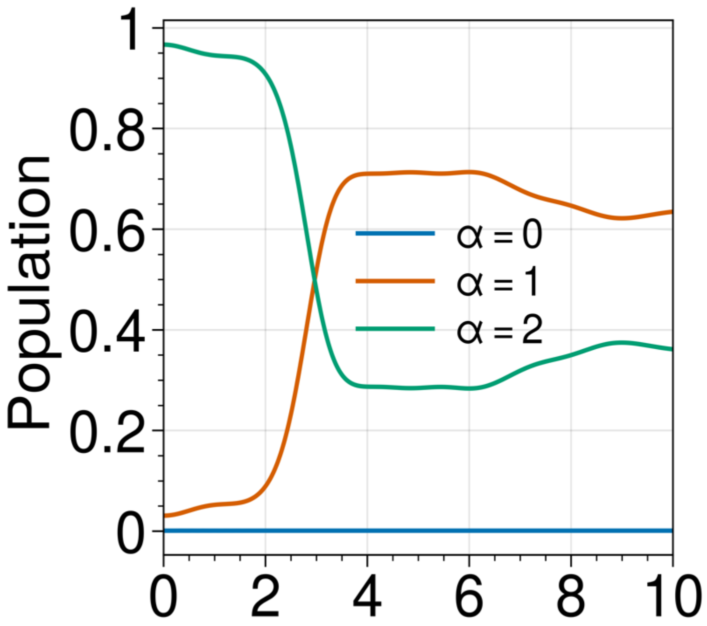


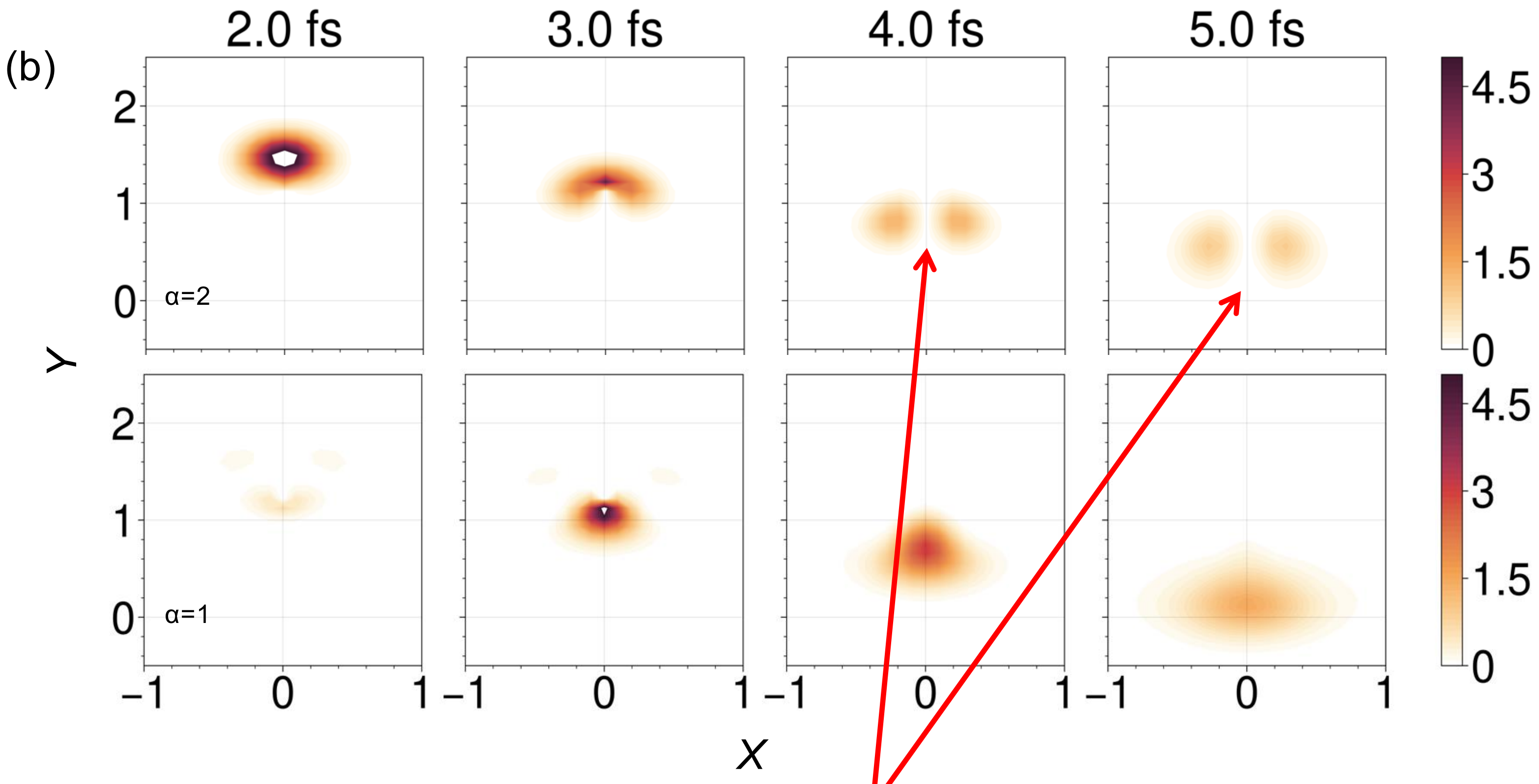


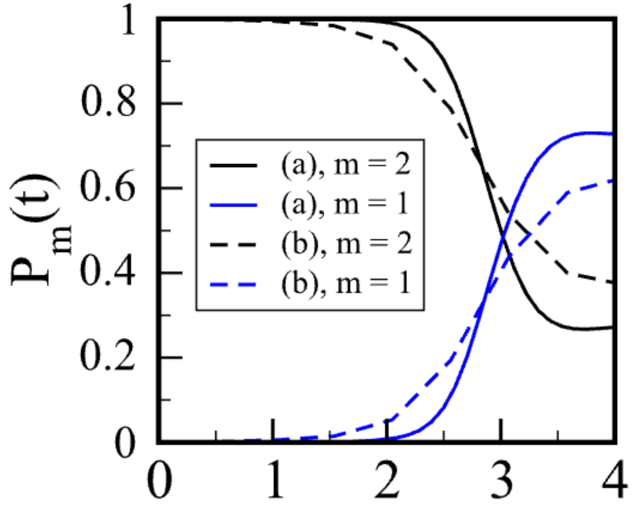


Zhu, X.; Gu, B. *J. Phys. Chem. Lett.* **2024**, *15* (33), 8487–8493

Ab initio LDR provides exact conical intersection dynamics



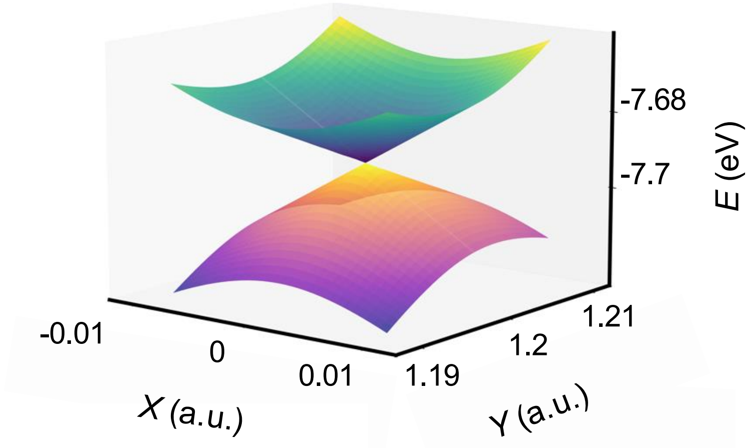


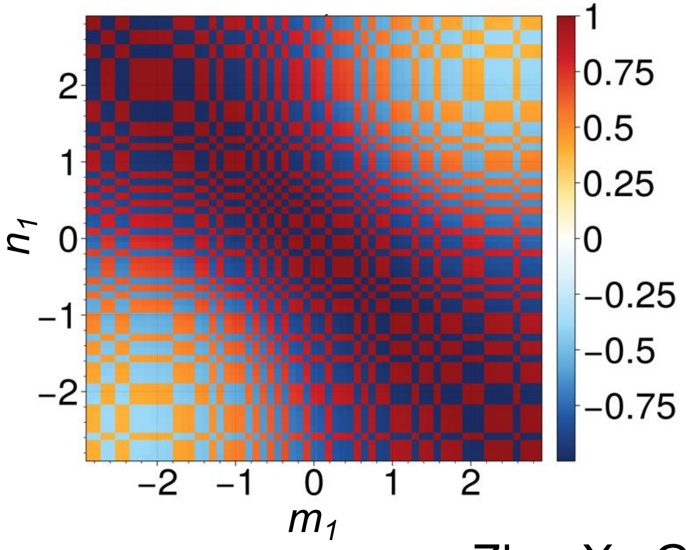


geometric phase manifested as a nodal line along X = 0

Hader et al. JCP **2017**, 146 (7), 074304

A more challenging case with an external magnetic field





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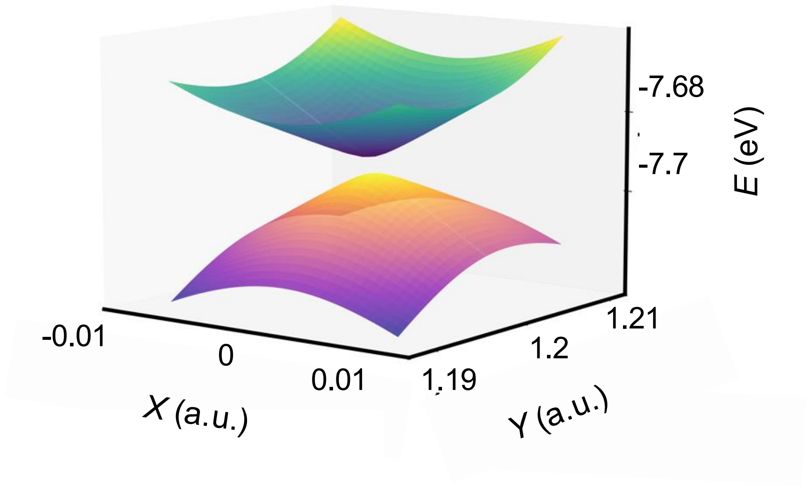
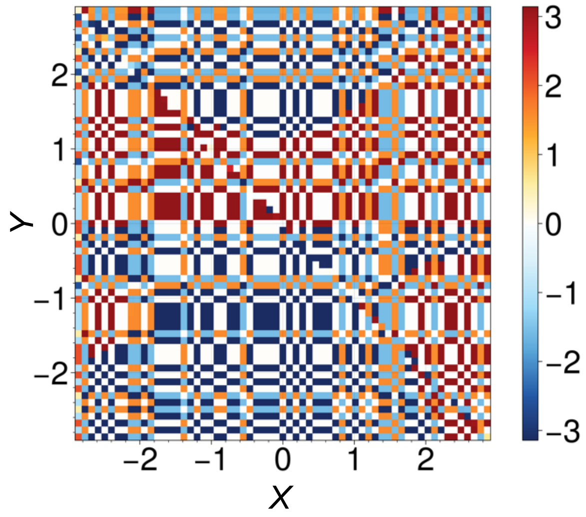
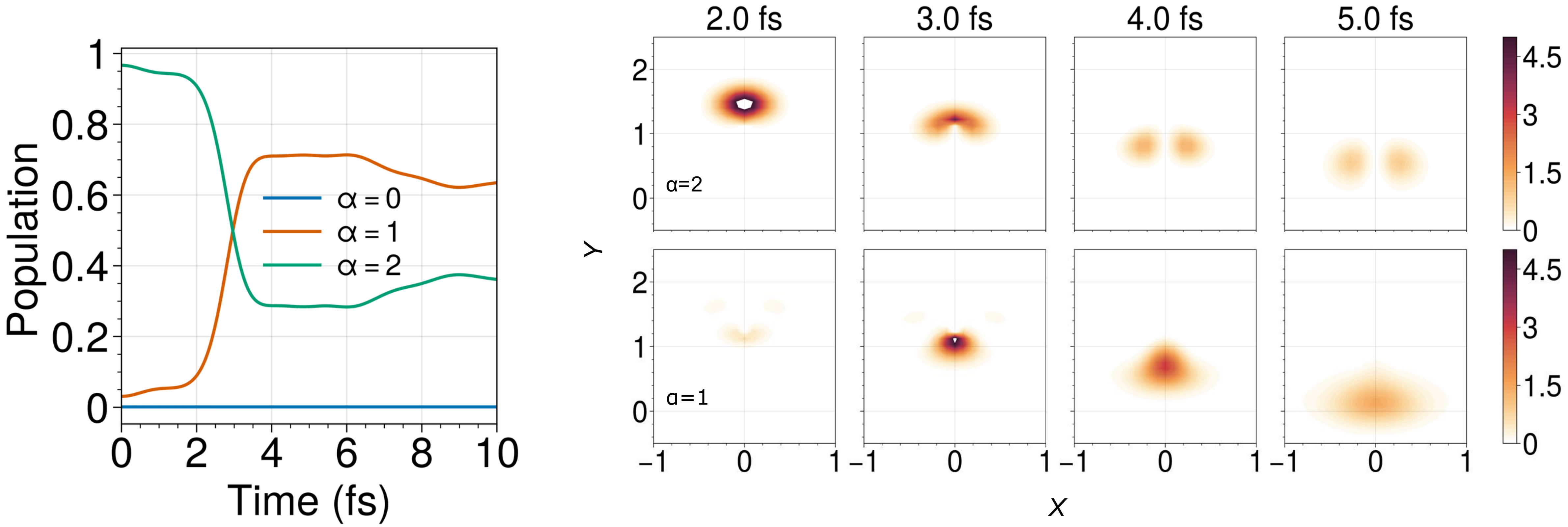


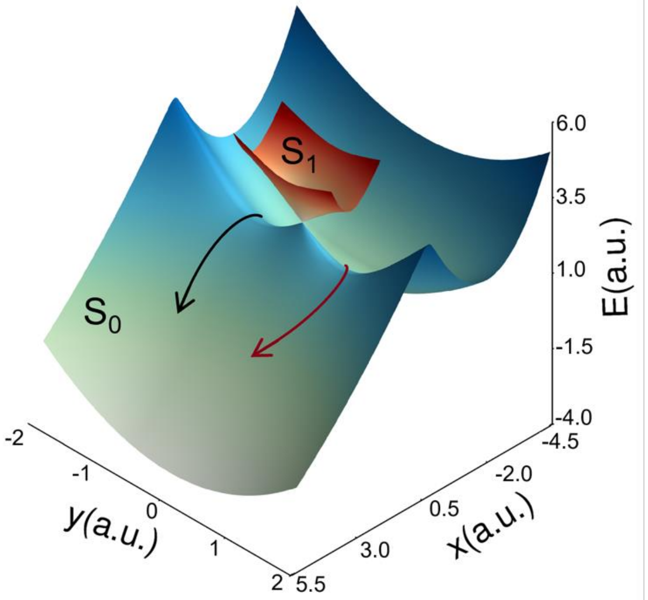


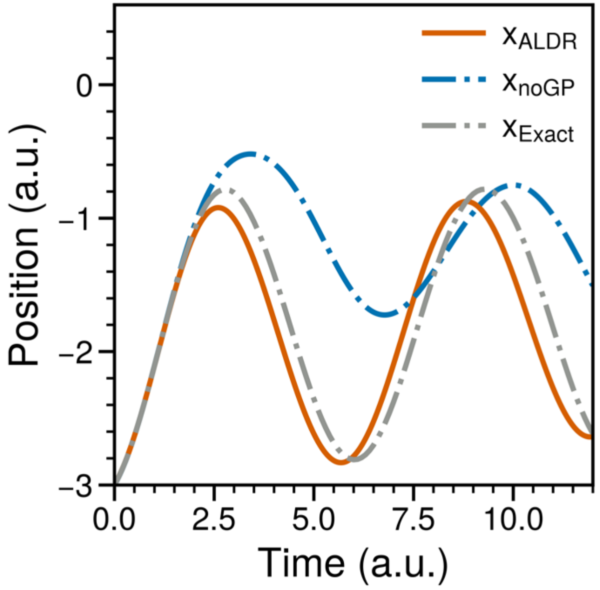
Zhu, X.; Gu, B. *J. Phys. Chem. Lett.* **2024**, *15* (33), 8487–8493

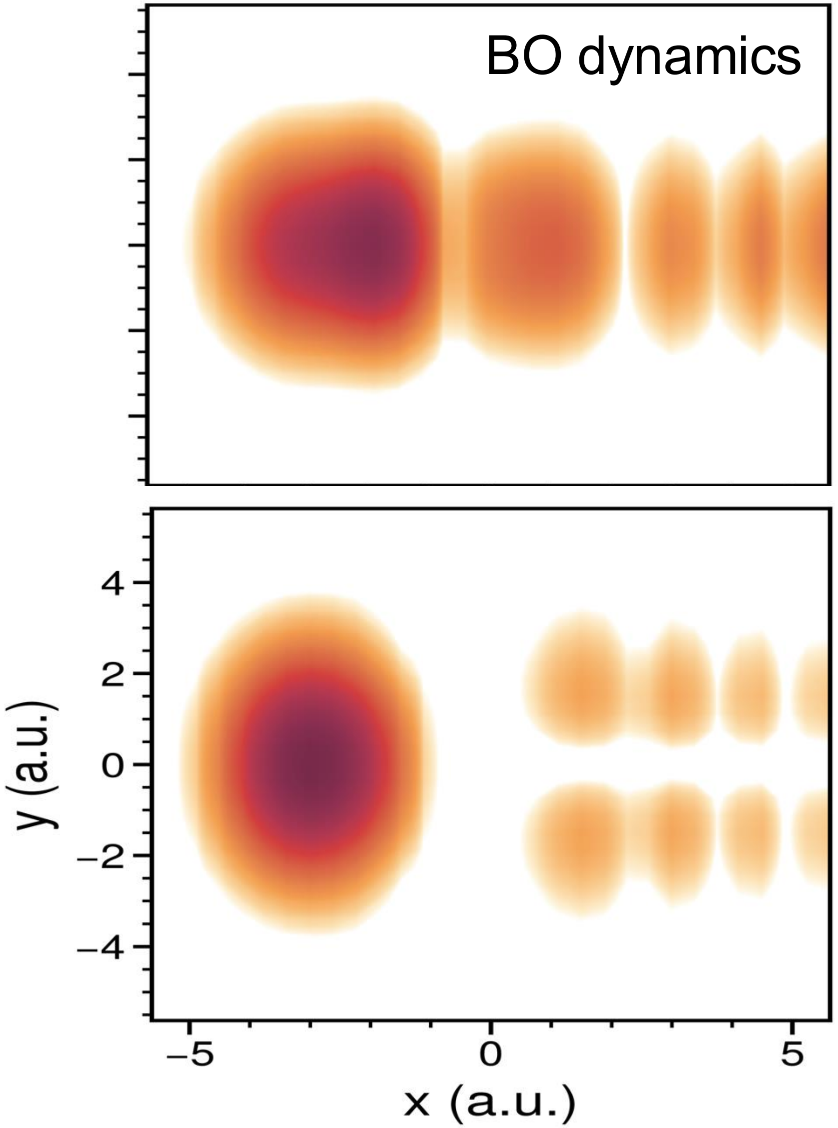


Zhu, X.; Gu, B. *J. Phys. Chem. Lett.* **2024**, *15* (33), 8487–8493

Model 4: Energetically Inaccessible Conical Intersections







Dr. Yujuan Xie

Second talk.

Why does it work?

What the overlap matrix encodes is quantum geometry

$$\ln \mathbf{A}(\mathbf{R}, \mathbf{R}+\boldsymbol{\Delta}) = \sum_{\mu} \mathbf{F}_{\mu}(\mathbf{R})\boldsymbol{\Delta}_{\mu} + \sum_{\mu,\nu} \frac{1}{2}\mathbf{Q}_{\mu\nu}(\mathbf{R})\boldsymbol{\Delta}_{\mu}\boldsymbol{\Delta}_{\nu} + O(\Delta^3)$$

$$F_{\mu}^{\beta\alpha}(\mathbf{R}) = \langle \phi_{\beta}(\mathbf{R}) | \partial_{\mu}\phi_{\alpha}(\mathbf{R}) \rangle \left\{ \begin{array}{l} \text{first-derivative coupling} \\ \text{vector potential} \end{array} \right.$$

$$Q_{\mu\nu}^{\beta\alpha}(\mathbf{R}) = \langle \partial_{\mu}\phi_{\beta}(\mathbf{R}) | 1 - \mathcal{P}(\mathbf{R}) | \partial_{\nu}\phi_{\alpha}(\mathbf{R}) \rangle$$

$$\mathcal{P}(\mathbf{R}) = \sum_{\alpha} |\phi_{\alpha}(\mathbf{R})\rangle\langle\phi_{\alpha}(\mathbf{R})|$$

Non-Abelian electronic quantum geometric tensor!

Y. Xie, R. Liu, B. Gu, *Topological Quantum Molecular Dynamics*, submitted

The **Abelian** quantum geometric tensor is the **central quantity** in topological physics

widely used among condensed matter physicists

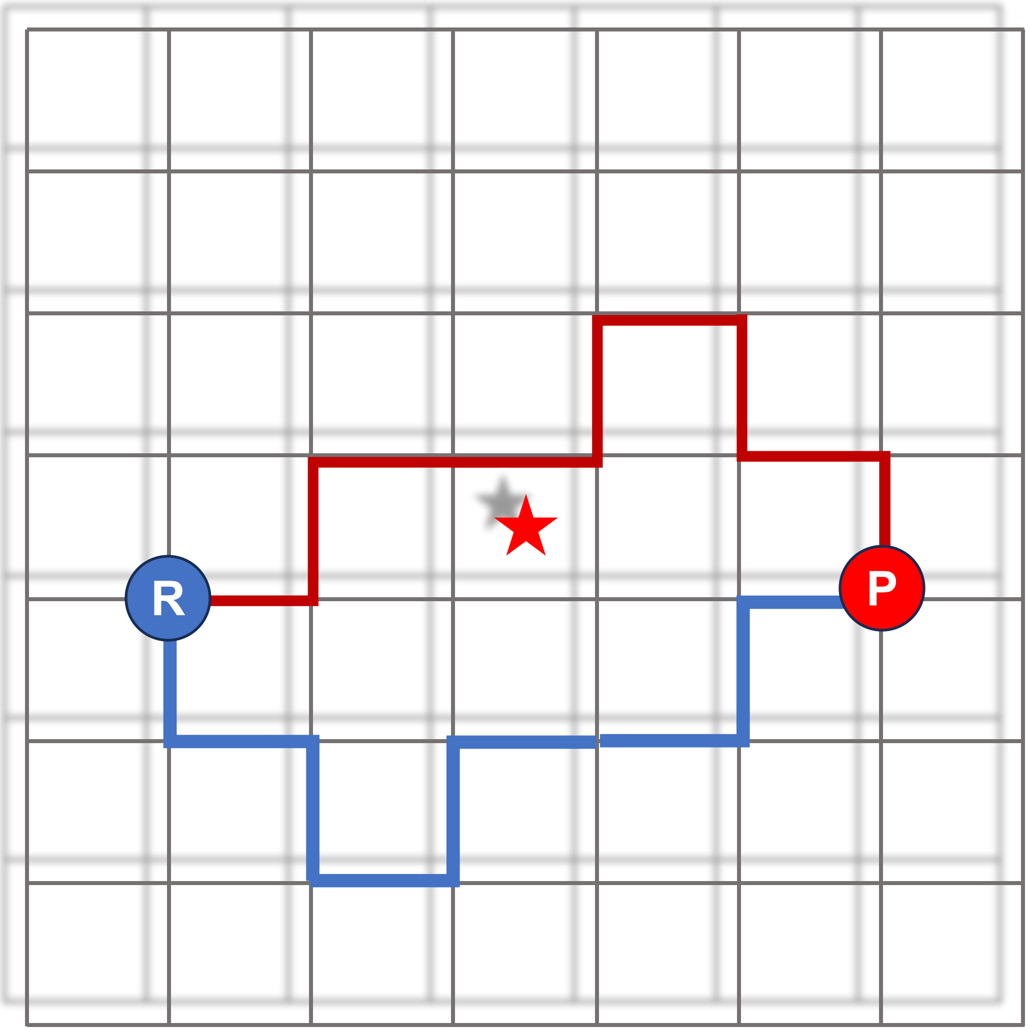
Likewise, the distances between quantum states depend on the geometry of the eigenstate space, and this is captured by the quantum geometric tensor (QGT) [2] (or Fubini-Study metric) $\mathcal{B}_{ij}(\mathbf{k})$:

$$\mathcal{B}_{ij}(\mathbf{k}) = \langle \partial_i u_{\mathbf{k}} | \partial_j u_{\mathbf{k}} \rangle - \langle \partial_i u_{\mathbf{k}} | u_{\mathbf{k}} \rangle \langle u_{\mathbf{k}} | \partial_j u_{\mathbf{k}} \rangle, \qquad (1)$$

where $\partial_i \equiv \partial/(\partial k_i)$ with $i = x, y$ and $u_{\mathbf{k}}$ is a wave function parametrized by a quantity $\mathbf{k}$ (which could be, for example, the lattice momentum). Its real part, the quantum metric $\mathrm{Re}\mathcal{B}_{ij}(\mathbf{k}) \equiv g_{ij}$ tells about the orthogonality, i.e., amplitude distance of quantum states under small changes. The last term of Eq. (1) is real due to normalization, thus the imaginary part of the QGT is $\mathrm{Im}\,\mathcal{B}_{ij}(\mathbf{k}) = -i(\langle \partial_i u_{\mathbf{k}} | \partial_j u_{\mathbf{k}} \rangle - \langle \partial_j u_{\mathbf{k}} | \partial_i u_{\mathbf{k}} \rangle)/2$. From this one can see that $\mathrm{Im}\mathcal{B}_{ij}(\mathbf{k})$ is the well-known Berry curvature (defined in vector form as $i\boldsymbol{\nabla}_{\mathbf{k}} \times \langle u_{\mathbf{k}} | \boldsymbol{\nabla}_{\mathbf{k}} u_{\mathbf{k}} \rangle$), which provides information about changes of the eigenstate phase (for more information see [3,4]). As the integral of the Berry curvature gives the Chern number, the QGT contains information about the system topology too. These concepts have been long known

P. Törmä, PRL 131, 240001 (2023)

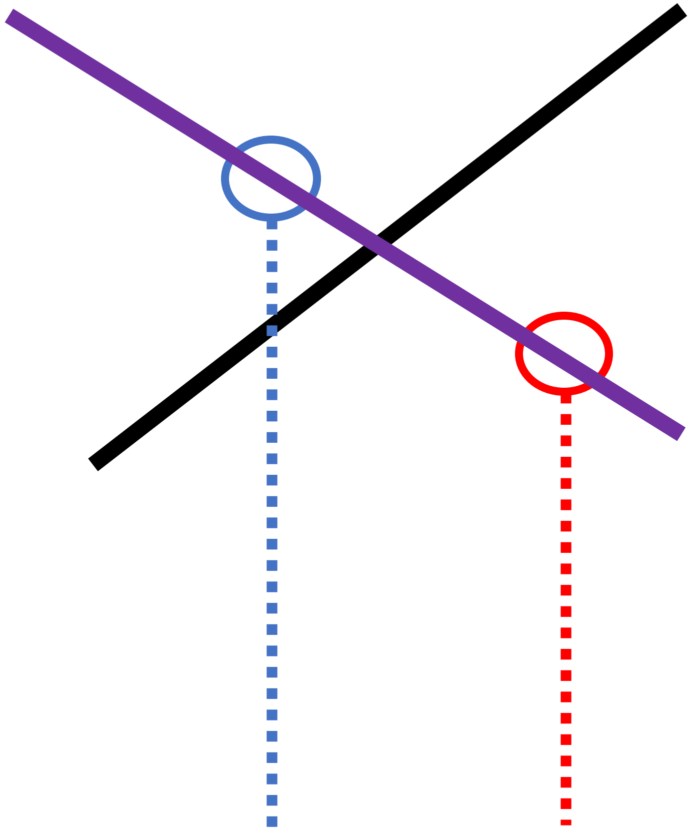
How is the geometric phase captured?



$$\text{relative phase} = \operatorname{sign} A(R_1, R_N) \cdots A(R_3, R_2) A(R_2, R_1) = -1$$

How does nonadiabatic transitions occur?

Electronic transitions occur by similarity



$$\langle \phi_e(\mathbf{R}) | \phi_g(\mathbf{R}') \rangle \sim 1$$

Diagonal Born-Oppenheimer corrections?

Contained (not equivalent) in the **diagonal matrix elements** of the electronic quantum geometric tensor

$$\mathrm{Re}\, Q^{\alpha\alpha}_{\mu\mu}(\mathbf{R}) = \mathrm{Re}\,\langle \partial_\mu \phi_\alpha(\mathbf{R}) | 1 - \mathcal{P}(\mathbf{R}) | \partial_\mu \phi_\alpha(\mathbf{R}) \rangle$$

DBOC does not appear alone

Summary

Electronic energies + Quantum Geometry = Exact Atomic Motion

A **universal** topological picture understanding molecular quantum dynamics – valid for all processes involving, e.g., doublets, triplets

- ✓ **Singularity-free**
- ✓ **Exact** – geometric phase, electronic coherence, nonadiabatic transitions
- ✓ Use **only** adiabatic electronic states
- ✓ Can deal with **a dense manifold** of electronic states

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Thank you!